DSA 210 Term Project Final Report

The Impact of Typhoon Signals on Hong Kong Stock Market Volatility and Returns

Hamza Eren İnan – 34502

Motivation

Financial markets are complex systems that continuously assimilate a vast spectrum of information into asset pricing dynamics, ranging from macroeconomic indicators and geopolitical shifts to extreme weather events. While the destructive impacts of natural disasters on physical infrastructure and local business operations are intuitive, how these physical disruptions map onto organized public equities and systemic volatility regimes remains an actively debated question in empirical finance. The primary motivation of this project is to explore how the official tropical cyclone warning mechanisms (Typhoon Signals) in Hong Kong—one of the world's premier global financial hubs—impact the short-term performance and behavior of its benchmark index, the Hang Seng Index (HSI). Due to its geographic location, Hong Kong is regularly subjected to severe typhoons every year. The warning signals issued by the Hong Kong Observatory (HKO), particularly Signal No. 8 and above, trigger structural emergency protocols across the city, leading to a temporary halt of public transit, business suspensions, and critically, the partial or complete closure of the Hong Kong Exchanges and Clearing Limited (HKEX). Whether such deterministic, scheduled trading halts and elevated climate risk perceptions introduce systematic behavioral biases among market participants, create a persistent directional anomaly in returns, or cluster volatility significantly upon market resumption is a crucial question for financial data science. This study adopts a data-driven framework to evaluate the market efficiency hypothesis and behavioral responses under acute weather shocks, leveraging statistical frameworks and machine learning models to map out trader behavior during predictable localized crises.

Data Source

To execute a reliable, multi-dimensional analysis, the core data infrastructure was established by enriching a public financial time-series with localized meteorological attributes across a unified temporal dimension:

Financial Dataset (HSI.csv): Sourced from public financial repositories on Kaggle, this comprehensive dataset covers end-of-day records for the Hang Seng Index spanning from 1986 to 2023, representingapproximately 7,500 trading days. The core continuous variables extracted include: Open, High, Low, Close, Adjusted Close prices, and daily trading Volume.

Meteorological Dataset (Typhoon.csv): Extracted from the historical database archives of the Hong Kong Observatory (HKO). This dataset logs approximately 1,200 unique tropical cyclone warnings and tracks details such as warning onset timestamps, typhoon taxonomy/names, warning signal strengths, and directional trajectories. Both data configurations were ingested locally as CSV records via Python's pandas library. To guarantee cross-dataset coherence and high structural fidelity, the primary analytical scope was restricted to the 2000– 2023 window, where the HKO warning records are completely structured, digital, and seamlessly map onto the active active market schedule.

Data Analysis

The data science pipeline implemented for this project is structured into five distinct phases: data cleaning, feature engineering, exploratory data analysis, parametric statistical testing, and the execution of supervised and unsupervised machine learning algorithms.

-Data Preprocessing & Cleaning

Chronological fields were standardized across both sheets into a synchronized YYYY-MM-DD baseline. Within Typhoon.csv, the "Strength" attribute presented mixed textual formats including localized Chinese directional characters (e.g., “8 東北” indicating Signal 8 Northeast). Regular Expressions (Regex) were utilized to parse these fields, isolating the underlying numeric alert level. Based on warning durations and institutional impacts, each distinct trading date was bucketed into three mutually exclusive operational risk categories:

-Normal Days (Category 0): Trading sessions with zero active typhoon warning signals.

-Low-Alert Days (Category 1): Days where Signal 1 or Signal 3 (standby and advisory notices) were active.

-High-Alert Days (Category 2): Trading dates matching severe Signal 8, 9, or 10 warnings, which trigger physical trading suspensions.

The processed meteorological observations were sequentially integrated into the HSI historical price series using a daily left-join approach. Unmatched trading dates were systematically imputed as "Normal" market sessions.

-Feature Engineering

To model market dynamics relative to weather alerts, the following continuous financial features were calculated from raw price points:

-Daily Return: Captures the percentage shift in closing values relative to the previous active session:

$$R_t = ((Close(t) - Close(t-1)) / Close(t-1)) \times 100$$

-Daily Volatility Metric: Reflects the intraday trading spread relative to the day's final close:

$$Volatility(t) = (High(t) - Low(t)) / Close(t)$$

-Trading Volume Change: The percentage delta in trading volume was computed to monitor liquidity anomalies.

-Exploratory Data Analysis (EDA) & Hypothesis Testing

Initial EDA revealed highly imbalanced class distributions, as normal trading sessions vastly outnumber typhoon-affected days. Due to the stark discrepancies in sample sizes and the potential variance heterogeneity between normal and crisis states, a traditional Student's t-test was abandoned in favor of Welch's t-test. The statistical testing design formulated a null hypothesis (H0) stating that the issuance of typhoon alerts produces no statistically significant difference in mean daily returns or underlying asset volatility compared to normal periods. The quantitative outcomes are summarized below:

Compared Target Environments	Tested Target Feature	Welch's t-Statistic	p-Value	Statistical Decision ( )
Normal vs. Low-Alert (Signals 1 & 3)	Daily Return	0.421	0.674	Fail to Reject H_0 (Insignificant)
Normal vs. High-Alert (Signals 8+)	Daily Return	-0.893	0.372	Fail to Reject H_0 (Insignificant)
Normal vs. High-Alert (Signals 8+)	Daily Volatility	1.105	0.270	Fail to Reject H_0 (Insignificant)

Because all computed p-values heavily exceeded the standard alpha threshold of 0.05, the null hypothesis could not be rejected. Statistically, the issuance of a typhoon signal does not generate an isolated, directional anomaly in daily returns or baseline index volatility.

-Machine Learning Methods

To supplement parametric tests, predictive and pattern-recognition models were introduced to assess whether complex non-linear combinations of market variables could classify typhoon states.

Supervised Learning (Classification): Daily Return, Volatility, and Volume Change were designated as the independent feature matrix, while the multi-class typhoon risk level acted as the classification target. Logistic Regression and Random Forest Classifiers were deployed, utilizing stratified data splits to protect model validation from severe class imbalances. Across all iterations, the testing accuracy consistently failed to outpace the naive baseline rate (the major class percentage). In essence, financial market attributes do not hold sufficient predictive information to reliably identify whether a severe weather alert is active on a given day.

Unsupervised Learning (Clustering): A **K-Means Clustering** routine was run blindly on the continuous financial attributes, completely omitting weather labels, to check if typhoon-affected days organically isolated into distinct market-regime clusters. The optimal layout partitioned the data into three clusters: (1) low- volatility, standard volume market conditions; (2) high-volatility, massive-volume macro reaction sessions; and (3) low-volume, stagnant trading windows. Mapping the historical typhoon labels onto these clusters showed that High-

Alert (Signal 8+) days were uniformly and proportionally distributed across all three clusters rather than pooling within the high-volatility cluster.

FINDINGS

The statistical tests and machine learning pipelines converged on several key analytical conclusions:

1-High Market Efficiency and Rapid Pricing: The Hang Seng Index exhibits strong-form efficiency regarding localized meteorological risks. The physical disruption caused by typhoons is processed and priced in virtually instantaneously, preventing any persistent directional trading anomalies from surfacing in the daily closing prices.

2-Absence of Structural Volatility Regimes: Even on severe Signal 8+ days that trigger full exchange shutdowns or truncated sessions, the overall daily volatility remains bound within the standard historical variance of normal trading days. Institutional traders, market makers, and automated algorithms proactively monitor HKO schedules, smoothing risk reallocations across adjacent trading days.

3-Indistinguishability at the Index Level: The uniform distribution of high-alert dates across unsupervised market clusters, coupled with the random-tier performance of classification models, underscores the resilience of Hong Kong's macro financial architecture against localized climate disruptions.

LIMITATIONS AND FUTURE WORK

Several core structural and data constraints boundary the findings of this research project:

Temporal Aggregation Mismatch: The analysis relies entirely on day-level data merging. However, typhoon alerts are intra-day phenomena that evolve on an hourly basis. Aggregating these dynamics into a single daily closing figure filters out short-term liquidity shocks, bid-ask spread expansions, or extreme panic/recovery waves that occur within a specific trading block.

Temporal Span of Overlapping Records: While HSI historical equity quotes date back to 1986, the structured, electronic HKO typhoon warning archive is only comprehensive from the year 2000 onward, narrowing the definitive analytical horizon.

Uncontrolled Confounding Macroeconomic Shocks: Major systemic events within the time frame (e.g., the 2008 Global Financial Crisis, Eurozone debt issues, the COVID-19 pandemic) were not explicitly introduced as control variables. Severe macro shocks overlapping with typhoon events could introduce unmodeled noise into the classifiers.

Recommendations for Future Work: Subsequent research should transition from daily metrics to high-frequency, intra-day tick data or 5-minute bar intervals. This granularity would allow for an explicit event-study analysis tracking order book dynamics, bid-ask spread expansions, and order cancellation rates immediately following HKO warning announcements. Furthermore, researchers should disaggregate the broad index into weather-sensitive sector portfolios—such as aviation, localized marine logistics, insurance, and real estate where physical weather impacts might reveal pronounced localized anomalies hidden at the macro index level.

Academic Integrity & AI Appendix

In accordance with Sabancı University's Academic Integrity Principles and the DSA 210 project guidelines, all external data sources utilized in this project have been appropriately cited. The overall structure of the project, the selection of the research topic, the formulation of hypotheses, and the data collection and core analytical design were conducted and originally developed entirely by me. During the project development process, artificial intelligence tools (Large Language Models) were utilized for support at specific stages. This assistance was strictly limited to the following areas:

Coding and Library Usage: AI assistance was used for syntax help when writing certain sections of the code, particularly for understanding and applying specific functions within data science libraries (such as pandas and scikit-learn).

Reporting and Formatting: While writing the final report, AI was utilized to help draft the text based on my analysis results, edit text templates, and assist with general document formatting.

The role of AI in this project was not to establish the research logic, conduct the core analysis, or interpret the scientific findings. Its use was solely to assist with technical implementation and accelerate the text editing process. All scientific conclusions, analytical insights, and intellectual property of this project belong entirely to me.